PARISHRAM 2.0

CLASS 12TH 2026

Botany-Sexual Reproduction in Flowering Plants

BIOLOGY

LECTURE-6

BY-AYUSHI AGGARWAL MA'AM



TOPICS TO BE COVERED

1. Agents of pollination
2. Pollen pistil interaction
3. Double Fertilization
4.

Pollination

Abiotic pollination ✓

(Wind & water)

Anemophily :- eg Poaceae

- Wheat
- Rice
- Bajra
- Jowar etc

↳ most common method of pollination among abiotic agent.

Adaptations

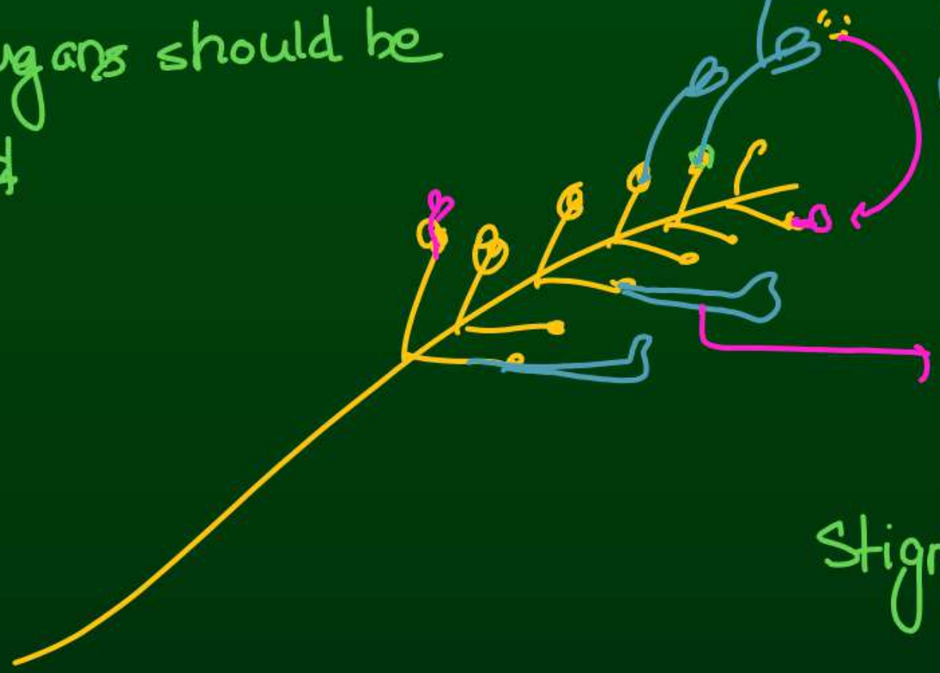
(i) Flowers are economical

i,e → non-brightly coloured
→ without fragrance
→ without nectar

Rather they are small, inconspicuous but borne in compact inflorescence → so that enormous quantity of pollengrain are produced.

(ii) Pollen grains are light & non-sticky

(iii) Sex organs should be exposed



filament long, so that anther is well-exposed

style & stigma should also be exposed to capture pollen grain

Stigma – sticky & feathery

(iv) Ovary has one ovule only

Hydrophily - ① Rarest medium for pollination because chance of successful pollination is low

② Only observed in 30 monocot genera.

③ Infacts water plants

- Lily
- Water hyacinth

→ Wind / Insect

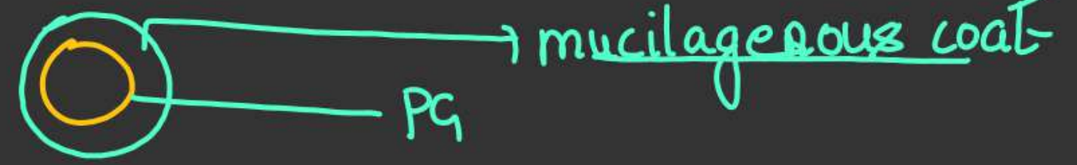
Hydrophily

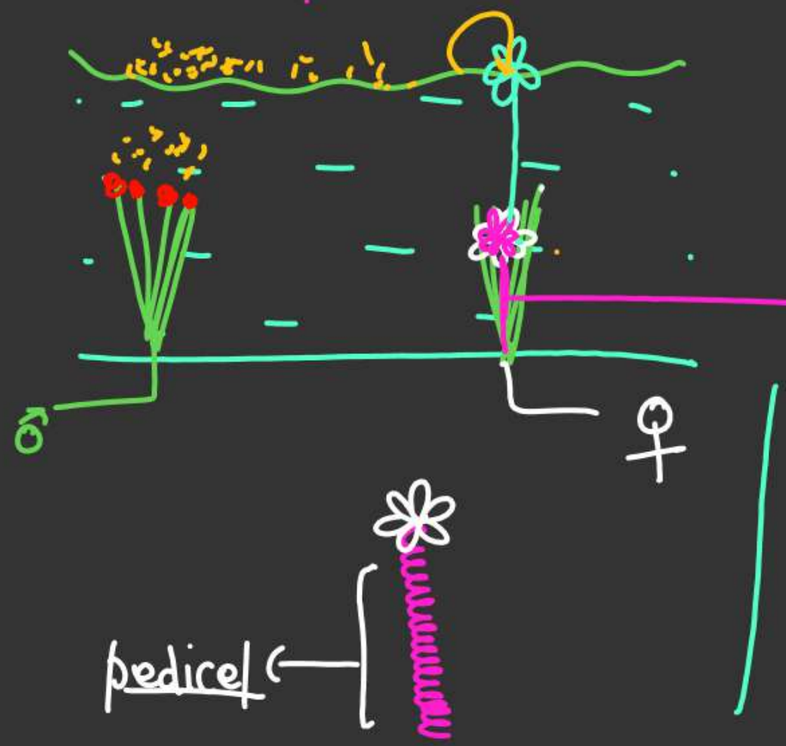
- Epihydro
  - eg Vallesneria (Tapegrass)
  - Dioecious
- Hypohydro
  - eg Zostera (Sea grass)

Epihydrophily

→ Pollen grain = light / non-sticky / non-wettable

For pollination, highly coiled pedicel opens, thus bringing the flower above the water.





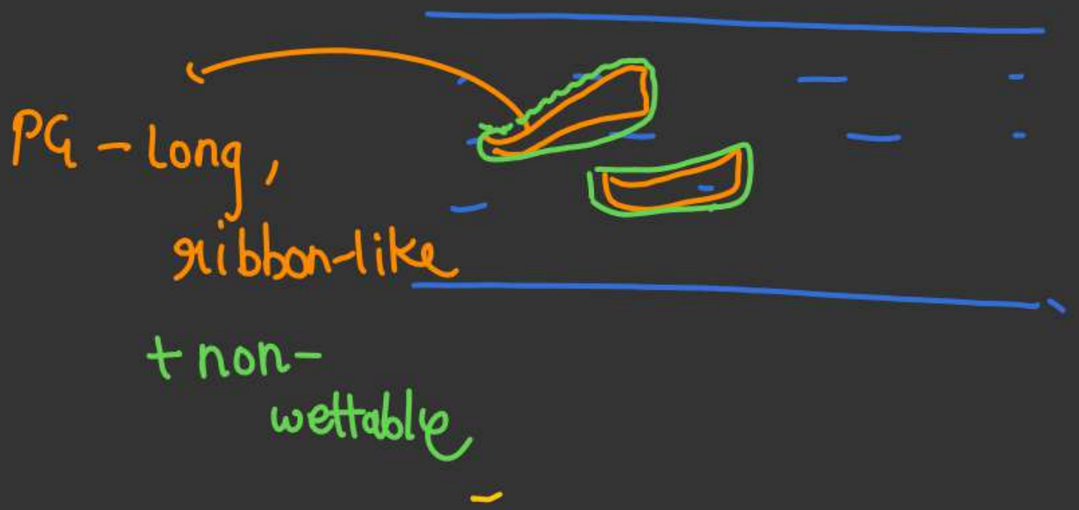
→ stalk of flower = pedicel = highly coiled

Agents of Pollination : Plants use two abiotic (wind and water) and one biotic (animals) agents to achieve pollination. Majority of plants use biotic agents for pollination. Only a small proportion of plants use abiotic agents. Pollen grains coming in contact with the stigma is a chance factor in both wind and water pollination. To compensate for this uncertainties and associated loss of pollen grains, the flowers produce enormous amount of pollen when compared to the number of ovules available for pollination.

Hydrophily :- pollination under water

↳ pollengrains donot reach the surface of water, rather swims inside the water.

eg Zostera



Biotic Agents

Entomophily → via insect (bee)

Plants provide benifits to its pollinator

- Safe place to lay egg
- Food
  - Nector
  - Pollen grain

+

Flowers are attractive

- Brightly coloured
- Large & showy
- full of fragrance

Pollination by wind is more common amongst abiotic pollinations. Wind pollination also requires that the pollen grains are light and non-sticky so that they can be transported in wind currents. They often possess well-exposed stamens (so that the pollens are easily dispersed into wind currents, Figure 1.10) and large often feathery stigma to easily trap air-borne pollen grains. Wind-pollinated flowers often have a single ovule in each ovary and numerous flowers packed into an inflorescence; a familiar example is the corn cob – the ears you see are nothing but the stigma and style which wave in the wind to trap pollen grains. Wind-pollination is quite common in grasses. Poaceae.

Majority of flowering plants use a range of animals as pollinating agents. Bees, butterflies, flies, beetles, wasps, ants, moths, birds (sunbirds and humming birds) and bats are the common pollinating agents. (Figure 1.11b). Among the animals, insects, particularly bees are the dominant biotic pollinating agents. Even larger animals such as some primates (lemurs), arboreal (tree-dwelling) rodents, or even reptiles (gecko lizard and garden lizard) have also been reported as pollinators in some species.

Often flowers of animal-pollinated plants are specifically adapted for a particular species of animal.

Majority of insect-pollinated flowers are large, colourful, fragrant and rich in nectar. When the flowers are small, a number of flowers are clustered into an inflorescence to make them conspicuous. Animals are attracted to flowers by colour and/or fragrance. The flowers pollinated by flies and beetles secrete foul odours to attract these animals. To sustain animal visits, the flowers have to provide rewards to the animals. Nectar and pollen grains are the usual floral rewards. For harvesting the reward(s) from the flower the animal visitor comes in contact with the anthers and the stigma. The body of the animal gets a coating of pollen grains, which are

Some insects like (flies, Beetles) → are attracted by the foul odour

eg

(i) Fig ⟷ Fig wasp.
Safe place to lay egg
pollination

(ii) Yucca / Amorphophallus ⟷ Moth
Safe place to lay egg
Pollination

Mutualism

generally sticky in animal pollinated flowers. When the animal carrying pollen on its body comes in contact with the stigma, it brings about pollination.

In some species floral rewards are in providing safe places to lay eggs; an example is that of the tallest flower of *Amorphophallus* (the flower itself is about 6 feet in height). A similar relationship exists between a species of moth and the plant *Yucca* where both species – moth and the

plant – cannot complete their life cycles without each other. The moth deposits its eggs in the locule of the ovary and the flower, in turn, gets pollinated by the moth. The larvae of the moth come out of the eggs as the seeds start developing.

3. Ophyrs plant & colpa moth

cheats the male colpa and attract it for pseudo copulation in return gets pollinated



QUESTION

Which of the following device is not used by plants to prevent autogamy?

1. Self - incompatibility
2. Production of unisexual flowers
3. Heterostyly
4. Production of cleistogamous flowers



QUESTION

Chasmogamy is pollination in -

open flowelled

1. Bud condition ✗
2. Closed flowers ✗
3. Unrelated flowers ✗
4. Opened flowers



QUESTION

Pollination in water hyacinth and water lily is brought about by the agency of:

1. Birds
2. Bats
3. Insects or wind
4. Water



QUESTION

Pollen grain of water pollinated plants are coated by covering to prevent it from wetting

1. mucilage
2. cuticle
3. exine
4. intine



QUESTION

The process whereby a perfect flower is pollinated by its pollen is called -

bisexual

1. Allogamy
2. Autogamy
3. Xenogamy
4. Hydrogamy



QUESTION

Which type of pollination occurs when pollen grains are transferred from the anther to the stigma of a different plant? (dioecious)

1. Autogamy
2. Xenogamy
3. Geitonogamy
4. Cleistogamy



QUESTION

What is the main advantage of cleistogamous flowers in terms of pollination?

1. They ensure assured seed-set without the need for pollinators.
2. They promote cross-pollination.
3. They attract more pollinators.
4. They produce fewer pollen grains.



QUESTION

Which of the following is not an agent of pollination?

1. Wind
2. Water
3. Insect and birds
4. None of these



QUESTION

Which of the following condition is helpful for wind pollination of flowers?

1. sticky pollens.
2. flowers with nectar.
3. well exposed stamens.
4. attractive colored petals.



QUESTION

Which of the following best describes geitonogamy in pollination?

1. Transfer of pollen grains between flowers of different plants
2. Transfer of pollen grains from the anther to the stigma of another flower of the same plant
3. Transfer of pollen grains from the anther to the stigma of a different species
4. Transfer of pollen grains within the same flower



THANK YOU